Meeting 7 of 7 · December 8, 2026

FINAL CAPSTONE DEFENSE

When is quantum error mitigation worth paying for?

A pre-registered study of selective error mitigation for noisy variational quantum fraud classifiers

Research problem: when should a noisy quantum classifier spend extra quantum resources on error mitigation, and is learning that decision better than a simple observable rule?

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Proposed split: Mayank (1–5, 10–11, 17–18) · Tushar (6–9, 12–16) · about 13–14 minutes

Sep 15 Proposal
Sep 29 Design
Oct 13 Implementation
Oct 27 Noise & mitigation
Nov 10 Policy selection
Nov 24 Final findings
Dec 8 Final defense

A noisy quantum classifier faces a spending decision

Readout mitigation

≈ 2 calibration circuits per batch

nearly free once calibrated

Zero-noise extrapolation

3× shots · 9× two-qubit-gate exposure

removes gate-noise bias

Extra shots

3× shots on the same circuit

removes variance, not bias

Why fraud scoring cares

Alerts fire at a fixed threshold, so biased probabilities raise the wrong alerts even when transactions are ranked well.

For every execution context: nothing, readout mitigation, ZNE, both, or more shots?

Source: p5c_costs.csv, configs/mitigation/*.yaml

What the literature offers

Mitigation foundations

- ZNE: Temme 2017 · Kandala 2019 · Giurgica-Tiron 2020 · Kim 2023
- Readout inversion: Bravyi 2021 · Maciejewski 2020
- PEC at exponential cost: van den Berg 2023
- Review: Cai 2023

Does mitigation pay?

- Beneficial on average, platform-dependent: Russo 2023
- At a matched shot budget, ZNE can be worse than raw: Zhang 2026
- Select by downstream decisions: Scavino 2026

Adaptive selection

- Telemetry + Thompson sampling: Szachara 2026
- Contextual bandit over ZNE scales: Rahman & Nguyen 2026
- Orchestration in quantum RL: Majid 2026
- Fuzzy rules on circuit features: Rao 2026

QML for fraud

- Same dataset; tuned classical baselines ahead: Dinuț 2026
- Architecture comparisons: El Alami 2026
- Better recall on bank data: Tudisco 2024
- Quantum kernels: Kyriienko 2022

Adaptive mitigation is not new; whether it beats simple rules was untested.

Source: full references on backup slide B6; verification in deliverables/paper/REFERENCES_VERIFICATION.md

Research gap

Work	Selects	Context used	Compared against
GSC-QEMit (Szachara et al. 2026)	abstract mitigation intensity	telemetry incl. ideal-state fidelity, simulator drift	unmitigated, static-severe
CMAB-ZNE (Rahman & Nguyen 2026)	ZNE scale-factor sets	loss / accuracy (label-dependent)	fixed and grid-search ZNE, other QEM
APGEM (Majid et al. 2026)	none / ZNE / PEC / CDR / readout	fidelity, reward, noise level	fixed techniques, Oracle
Rao et al. 2026	none / DD / ZNE (fuzzy rules)	label-free circuit + hardware features	not stated in abstract
This work	**none / readout / ZNE / both / extra shots**	**label-free probes only (leakage-tested)**	**simple rules + learned, held-out, pre-registered**

1 **Extra shots and combined mitigation in one action space**

2 **Label-free probes only: no labels, no simulator parameters**

3 **Learning vs simple rules on held-out models, pre-registered**

Gap: a pre-registered, held-out test of learned selection against simple observable rules.

Source: deliverables/paper/REFERENCES_VERIFICATION.md

Original hypothesis and how we tested it

H-control (original)

A learned controller using only label-free probes beats fixed policies and a fairly tuned threshold rule.

The alternative we committed to report

A simple threshold rule matches the learned controller; formalized for the sealed test as H1–H4.

Gate G3B
categorical noise: residual for learning?

→

Gate G3C
continuous joint noise: residual for learning?

→

Sealed test
opened once: H1–H4

H1 rule near-Oracle: material failure ≤ 0.05, Brier regret ≤ 2%

H2 no learned comparator meaningfully better (non-superiority)

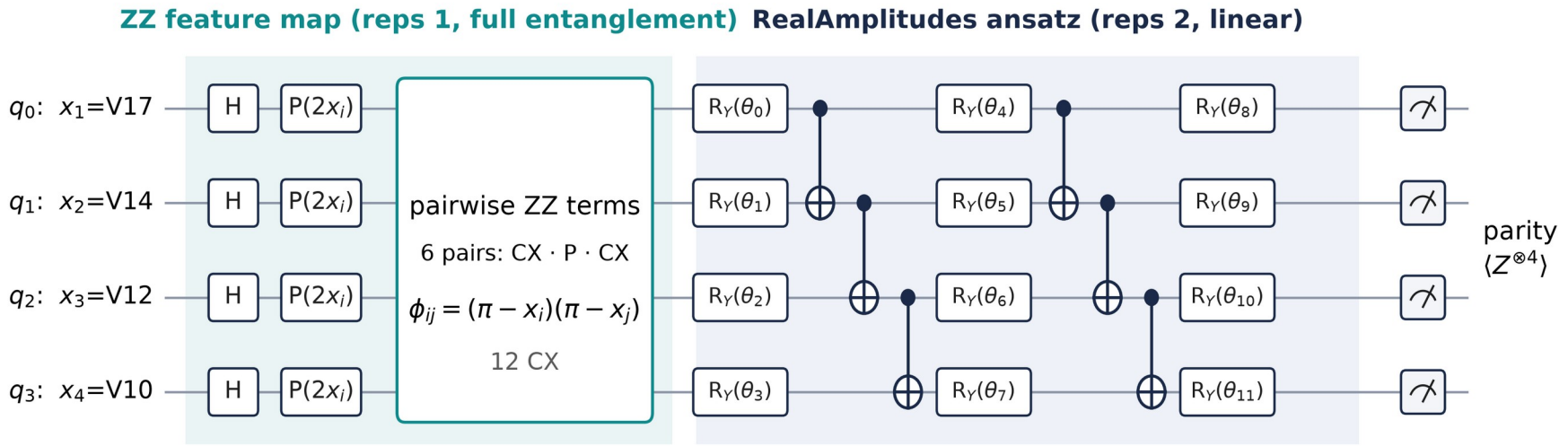
H3 rule beats every universal policy, ≤ always-A3 shots

H4 rule transfers to categorical noise: material ≤ 0.10

Negative outcomes were registered as reportable results before any data existed.

Source: docs/PREREG_P5B.md, docs/PREREG_P5C.md, docs/PREREG_FINAL.md, configs/final_test/final_v1.yaml

QML fraud-classification architecture



$p(\text{fraud}) = (1 - \langle Z^{\otimes 4} \rangle)/2$ · encoding e3: clip ±15 IQR → $x_i \in [-\pi/4, \pi/4]$ · 12 trainable parameters · 18 CX after transpilation

0.104 → 0.507

validation AUPRC after the pre-registered encoding repair (AUROC 0.957)

0.59–0.62

classical AUPRC on the same 4 inputs: no quantum advantage

64 variants

16 structures × 4 seeds for the benchmark (slide 12)

The VQC is the noisy workload under study, not a competitor to classical fraud models.

Source: configs/repair/p3ar_v1.yaml, p5a_pool_metrics.csv, p3ar_validation_comparison.csv

Dataset and leakage-safe preprocessing

284,807 transactions
492 frauds

→

Exact deduplication
1,081 duplicate rows removed (773 groups)

→

283,726 rows
473 frauds · 0.17%

→

Group-aware split
0 groups cross partitions

Partition	Rows	Frauds	Used for
VQC train	127,677	213	fitting every VQC and the preprocessing
VQC validation	28,372	47	qualification, thresholds, characterization
Controller train	42,559	71	policy fitting (training models)
Controller validation	28,372	47	policy validation (validation models)
Controller test	56,746	95	**sealed; opened once (test models)**

Leakage controls

- Preprocessing fitted on VQC train only
- Offline-only: true noise, labels, IDs
- Feature registry + automated tests

Five disjoint, group-aware partitions; the controller test stayed sealed until the end.

Source: ulb_v1_dataset_summary.csv, ulb_v1_phase1_metadata.json

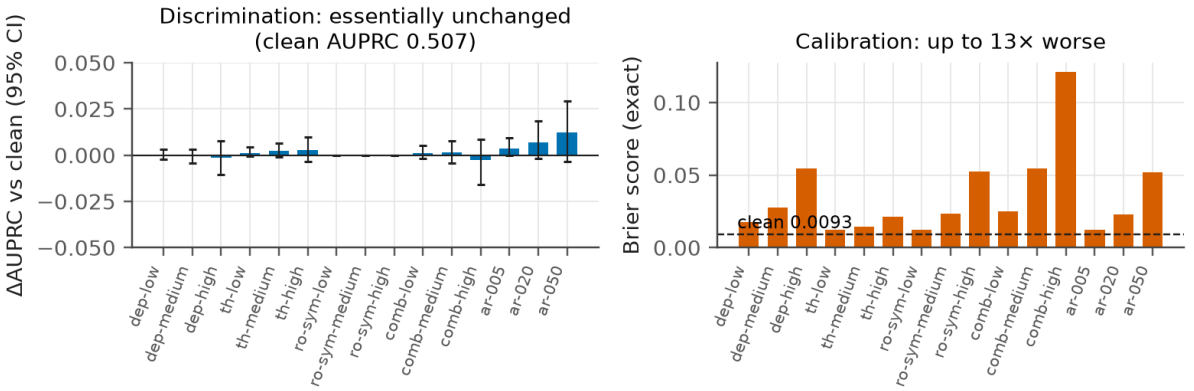
Noise models

Depolarizing
two-qubit error 0.005–0.020

Thermal relaxation
T_1 200 → 50 µs

Readout confusion
flip probability 0.005–0.05

Noise reshapes calibration, not discrimination (P3A-R on vqc_val; Phase 3B)



≤ 0.003
change in AUPRC

13×
worse Brier (0.0093 → 0.121)

5B: 8 near-categorical conditions · 5C: 17 gate settings (one gate-free) × 4 readout = 68 points

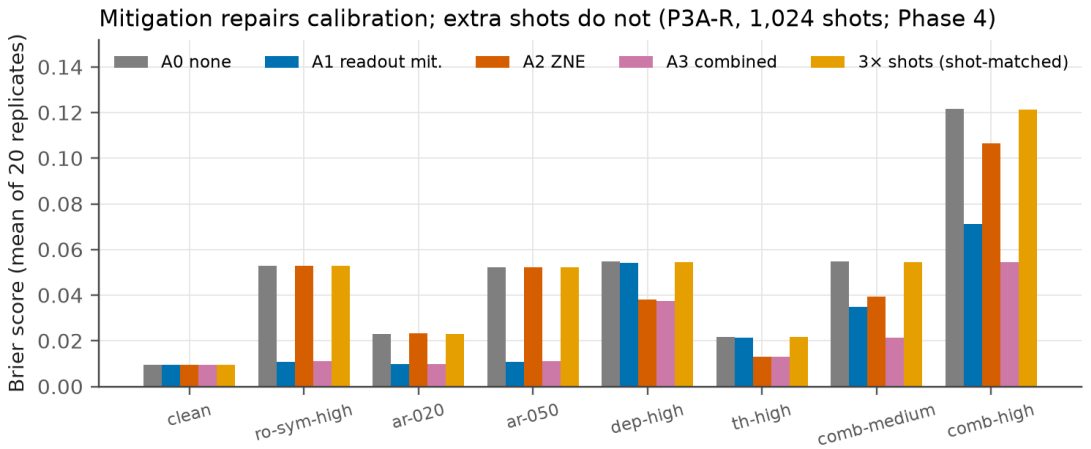
Noise damaged calibration, not discrimination.

Source: configs/noise/anchors_rr_v1.yaml, p5c_design.csv, p3b_exact_metrics.csv

Error-mitigation methods

Action	Method	Shots / inference	CX exposure
A0	none	S	1×
A1	tensored readout inversion + simplex projection	S + 2S/N	1×
A4	3× shots, unmitigated (resource-matched)	3S	3×
A2	ZNE: global folding λ = 1, 3, 5; linear fit	3S	9×
A3	readout mitigation + ZNE	3S + 2S/N	9×

S = shots per circuit · N = transactions per batch



0.122 → 0.054

Brier, combined-high, combined mitigation

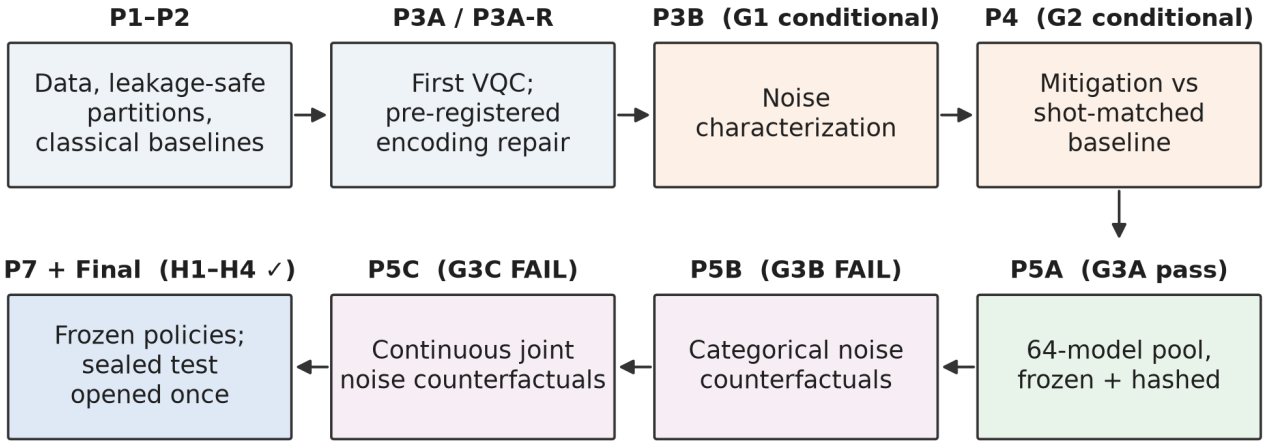
0.121

Brier with 3× shots: unchanged

Mitigation removes a bias that shots cannot; which action pays depends on the noise.

Source: p4_summary.csv (1,024 shots), p5c_costs.csv

Experimental design and integrity



From Phase 3A-R on, each phase was registered and committed before its own outcomes; reported runs used a clean tree.

Gates decided whether to continue; controller-test outcomes stayed sealed until the final protocol was pushed.

Pre-registered
from 3A-R on, before each phase's outcomes

Sealed test
one guarded unlock; reads logged

Frozen comparators
decision fingerprints verified

Reproducible
16 slow tests re-run Phases 1–5C + final

Gates decided whether to proceed; negative outcomes were reported as results.

Source: docs/PREREG_*.md, final_metadata.json, tests/

Evaluation metrics

Primary endpoint

Brier score of the mitigated fraud probabilities, mean of 20 finite-shot replicates

Oracle p*

cheapest action whose Brier score is within ε = 10% of the best (evaluation only)

Failure

any disagreement with p*

Material failure

Brier > (1 + 2ε) × best, or ≥ 1 extra shot-set (S) per inference

Cost

shots per inference, including 66·S probe shots per batch for probing policies

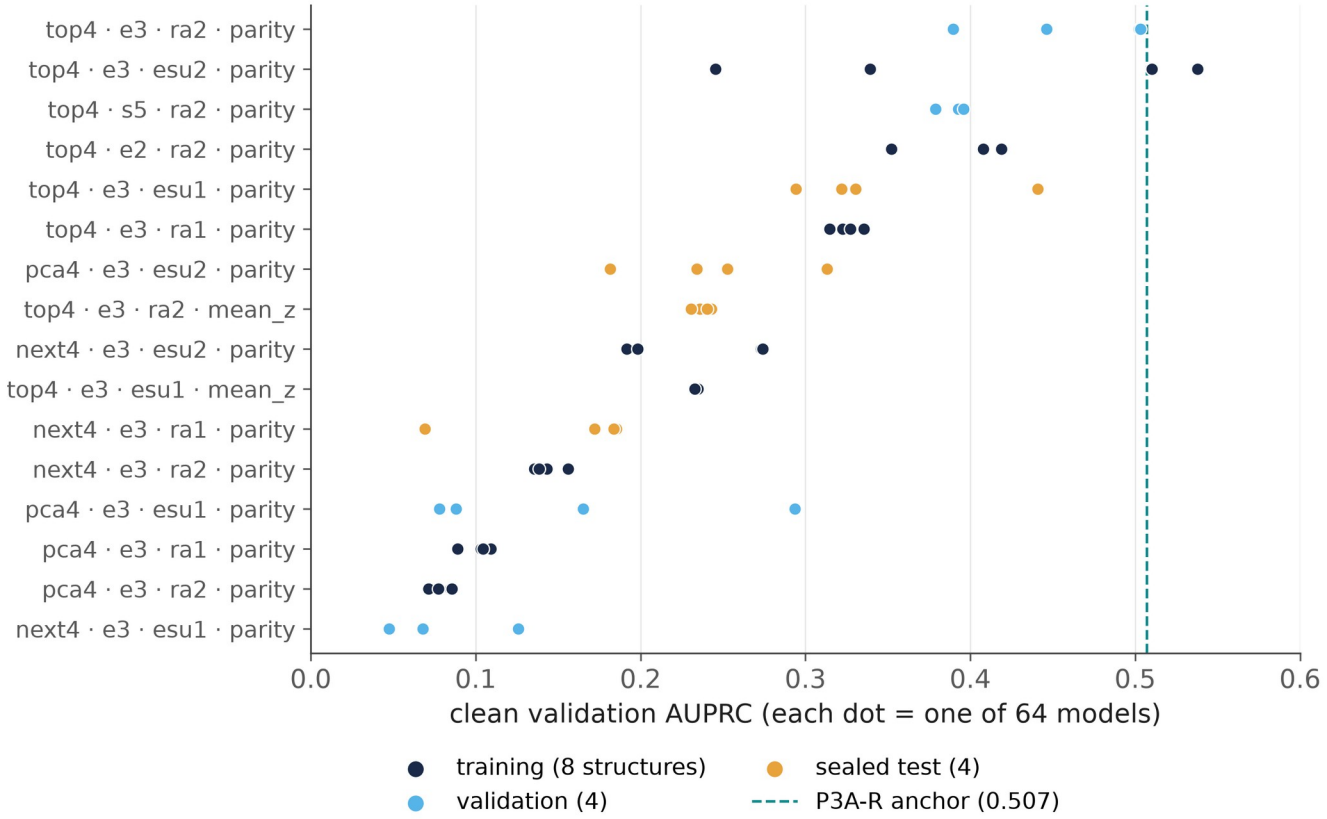
Uncertainty

two-stage cluster bootstrap: structures, then models (2,000 resamples)

Brier regret can be slightly negative: the Oracle may trade up to 10% of reliability for a lower cost.

Source: configs/mixed_noise_study/p5c_v1.yaml, configs/final_test/final_v1.yaml

The 64-model benchmark



64 models

16 structures × 4 seeds; median clean AUPRC 0.235 (0.047–0.538)

≈ 1.4 million

paired counterfactual outcomes (5B 102,400; 5C 1,305,600)

66 circuits

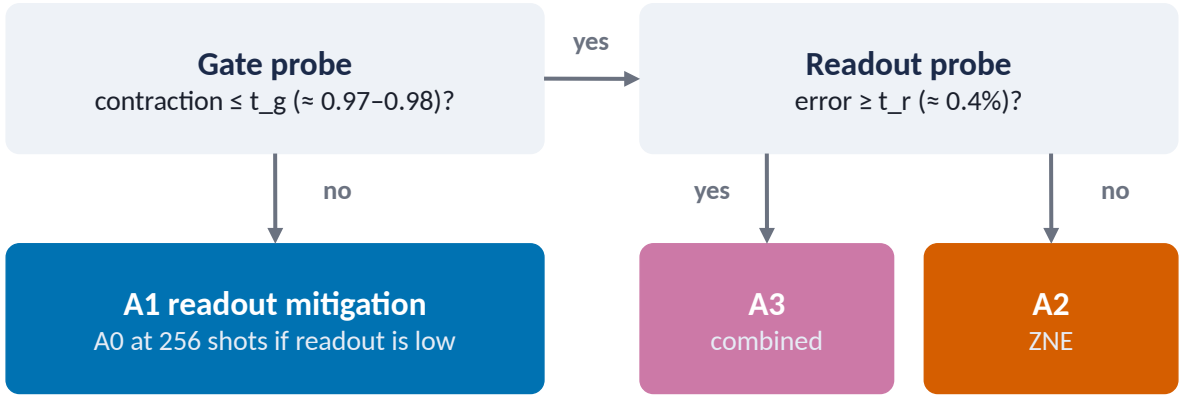
probe cost per batch: 2 readout calibration + 32 gate bindings × folds 1, 3

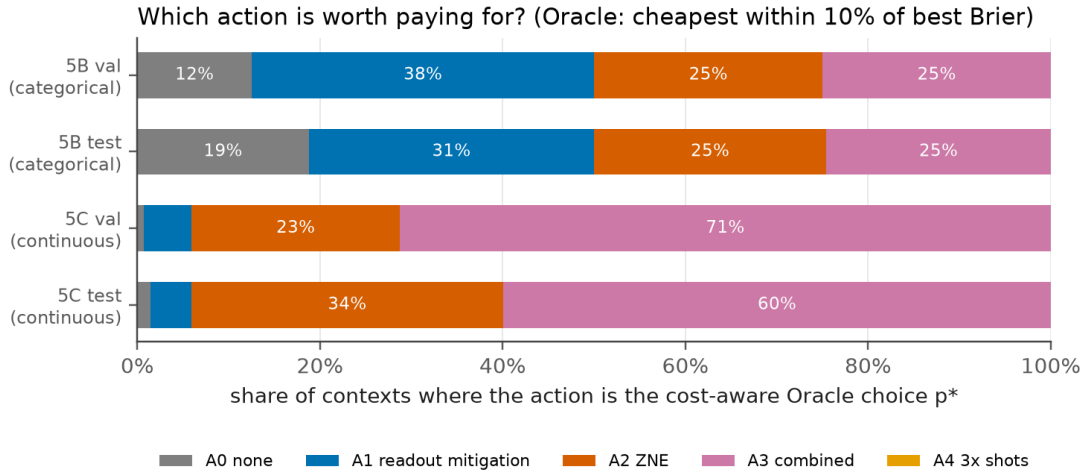
Models, nested in structures, are the independent units; 4 test structures stayed sealed.

Source: p5a_pool_metrics.csv, p5b_split.csv, p5b_feature_definitions.csv; row counts per docs/EXPERIMENT_PROTOCOL.md §3.12–3.13

Policy comparison: the gates failed, a simple rule emerged

The probe rule R_thr





G3B → FAIL

category rule wrong in 0.33% of validation cases

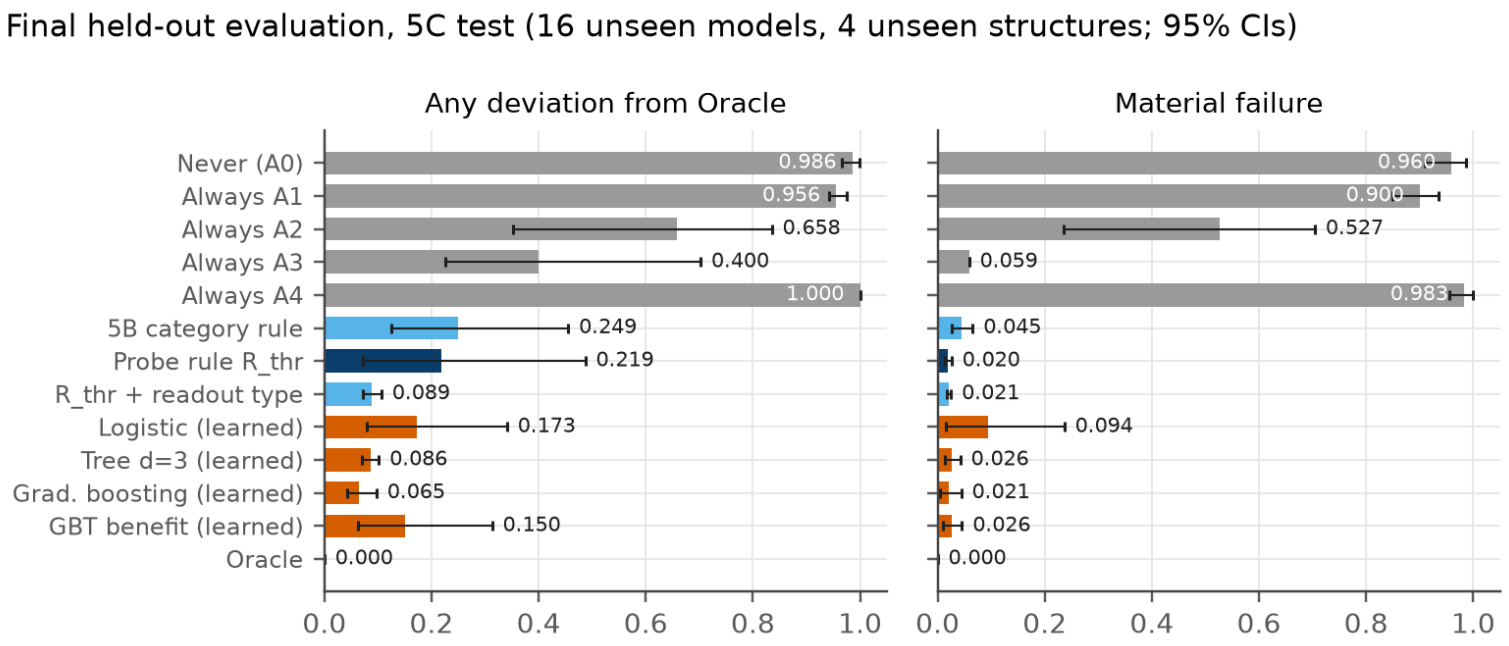
G3C → FAIL

R_thr material failure 2.0% [1.6, 2.7]

As registered, no learned controller was trained; fixed learned comparators were tested instead.

Source: p5b_metadata.json, p5c_metadata.json, final_metadata.json

Final held-out results



2.0%

material errors, rule (95% CI 1.4–2.6%; 4 structures, likely too narrow)

2.1%

best learned (gradient boosting): a tie

8.8%

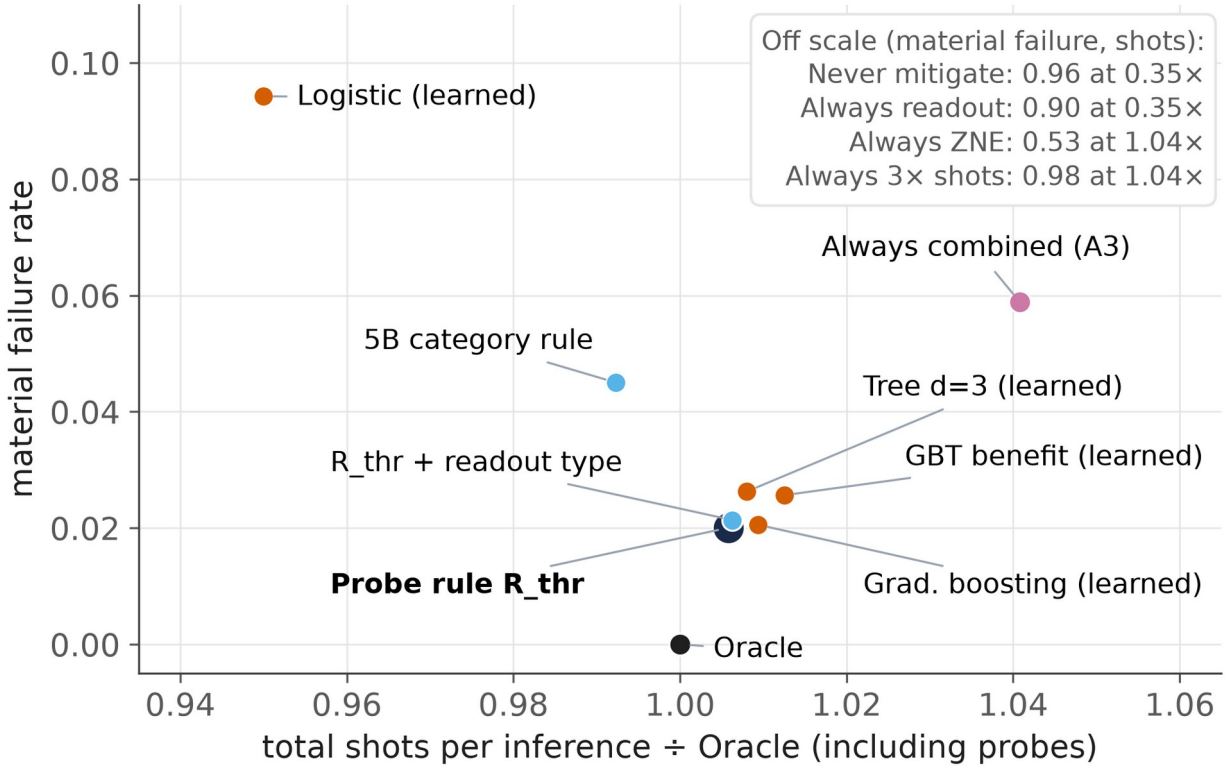
rule, categorical test (bound 10%)

H1 ✓ H2 ✓ H3 ✓ H4 ✓

Any disagreement: gradient boosting 6.5% vs rule 21.9%, mostly on mean-Z, where the rule's extra choices were cheap.

Source: final_policies_5c_test.csv, final_policies_5b_test.csv, final_paired_vs_rthr_5c_test.csv, final_metadata.json

Cost–reliability findings



+0.6%

shots vs the Oracle (rule)

−3.4%

shots vs always-combined

0.35× · 96%

never mitigate: shots · material errors

Always-combined's 5.9% = 1 gate-free setting in 17

The saving over always-combined depends on how often batches are gate-free.

Source: final_policies_5c_test.csv, p5c_design.csv

Limitations and threats to validity

Simulation only
no hardware run; asymmetric readout synthetic; no crosstalk, drift or coherent errors

Small scale
4 qubits, shallow circuits, one public dataset

No quantum advantage
classical models score higher on the same inputs

Narrow mitigation set
readout inversion and linear ZNE only; no PEC, CDR or DD

Few held-out structures
only 4: intervals likely too narrow; H2 is non-superiority

Probe, design and endpoint
thermal blind spot; 16 of 17 gate settings noisy; learned better on raw agreement

Source: final_policies_5c_test.csv, final_paired_vs_rthr_5c_test.csv, p5c_probe_validity.csv, p3ar_validation_comparison.csv

Research contribution

Characterization

Noise damaged calibration, not discrimination; mitigation repaired calibration; extra shots did not.

Benchmark

Leakage-safe counterfactual benchmark: 64 VQCs × noise × budget × 5 actions, ≈ 1.4 M outcomes.

Pre-registered test

A held-out, one-time test of learned selection against simple rules: the rule was competitive.

Artifacts: manuscript (LaTeX), capstone report, pre-registrations, code and aggregate tables, 16 reproduction tests. Nothing has been submitted.

Evaluate adaptive mitigation against simple probe rules and a cost-aware Oracle, on held-out models.

Source: deliverables/README.md

Conclusions and future work

What we found

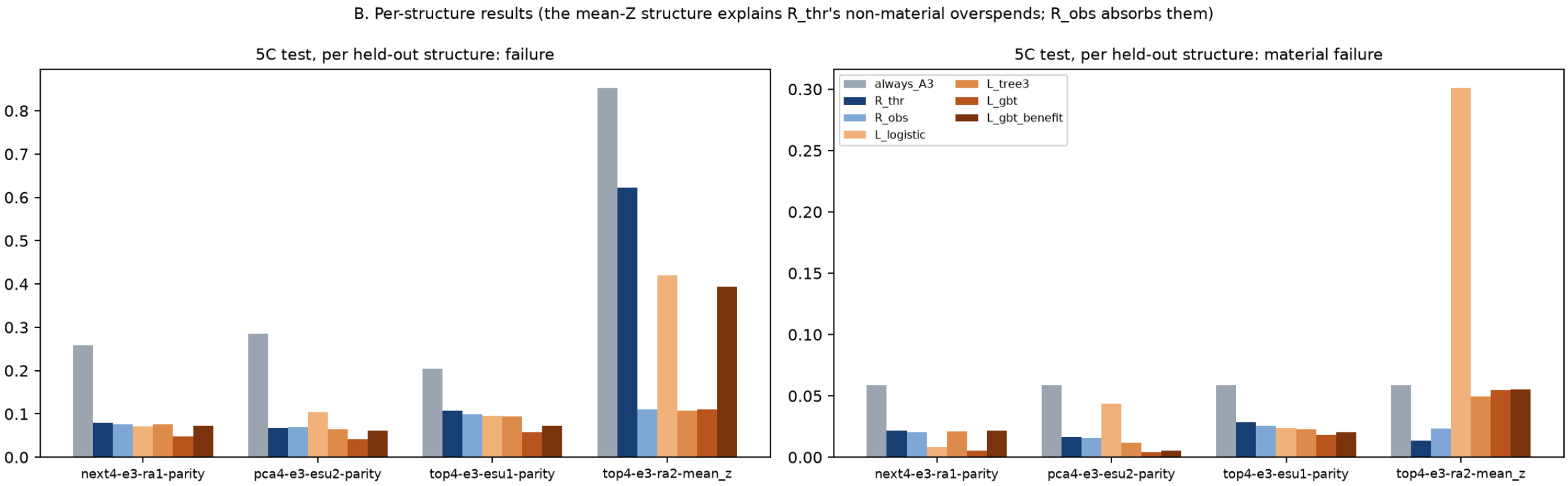
- Mitigation fixes calibration, not discrimination; extra shots never substitute.
- In our joint-noise design the material question is: is gate noise present?
- A two-threshold probe rule was near-Oracle on unseen models; fixed learned comparators did not reduce material errors.

Future research

- Hardware validation with real readout asymmetry, drift and crosstalk.
- Richer mitigation (PEC, CDR, DD) and a thermal-sensitive gate probe.
- Larger circuits, where noise may also degrade discrimination.
- Endpoints that weight fraud-operations costs.

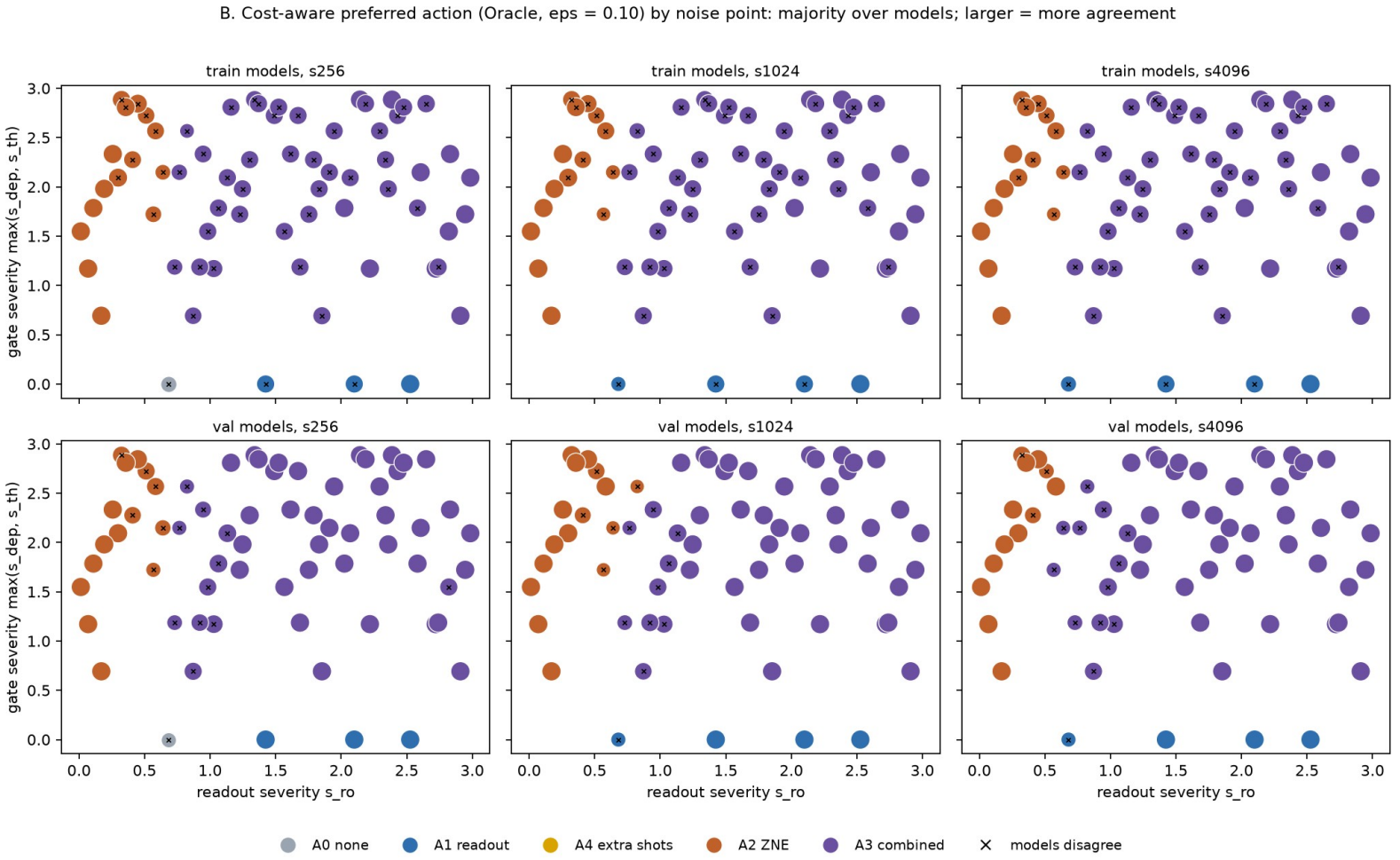
In the studied setting, a simple observable probe rule performed competitively with learned comparators.

B1 · Per-structure held-out results



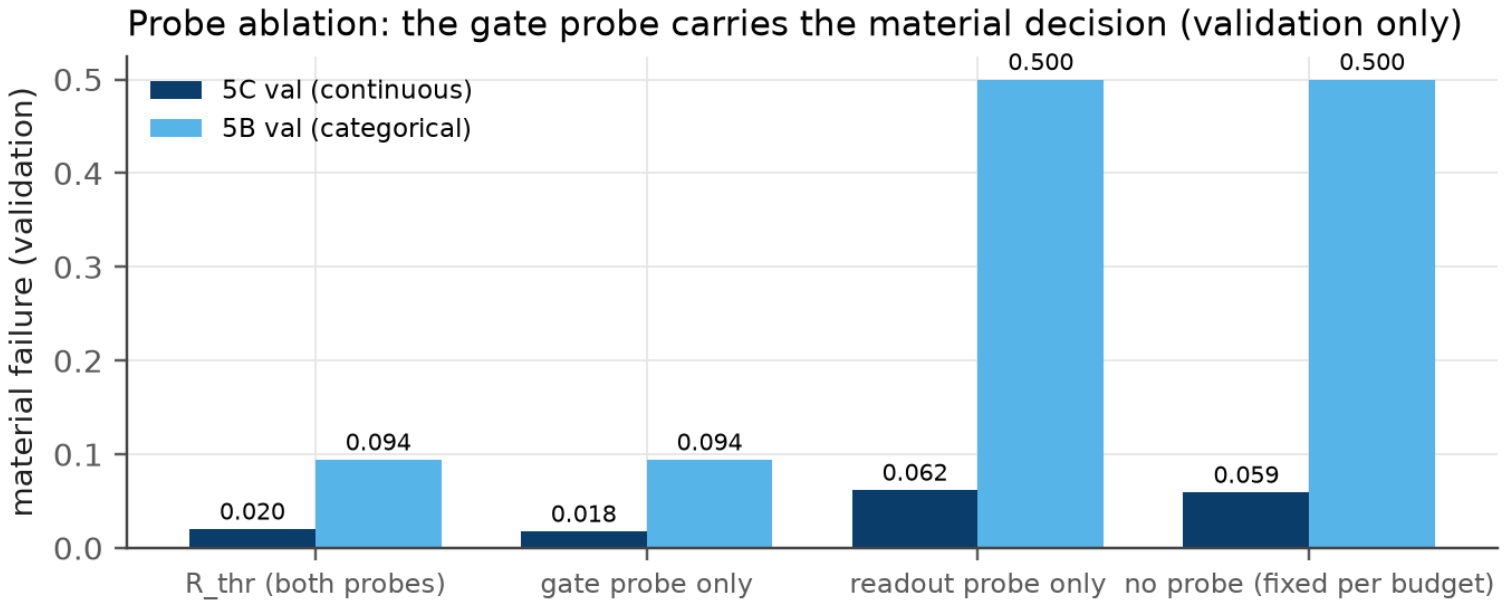
Source: final_by_structure_5c_test.csv (results/figures/final_b_structures.png)

B2 · Where ZNE versus combined mitigation wins



Source: p5c_preferred_by_point_train.csv, p5c_preferred_by_point_val.csv

B3 · Probe ablation and probing break-even



1.8%

material failure, gate probe alone (5C validation)

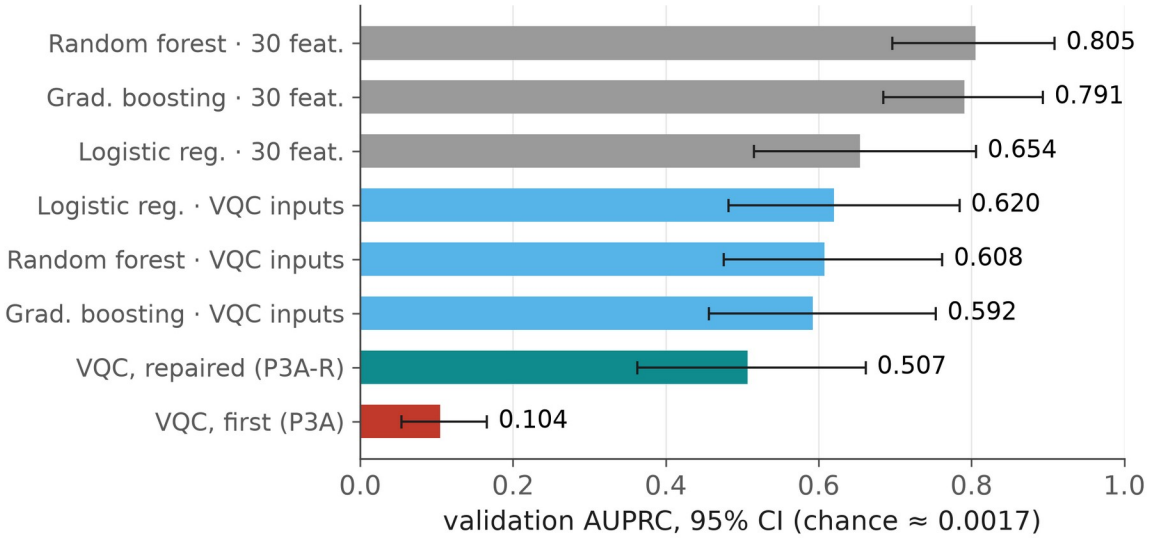
≈ 645–685

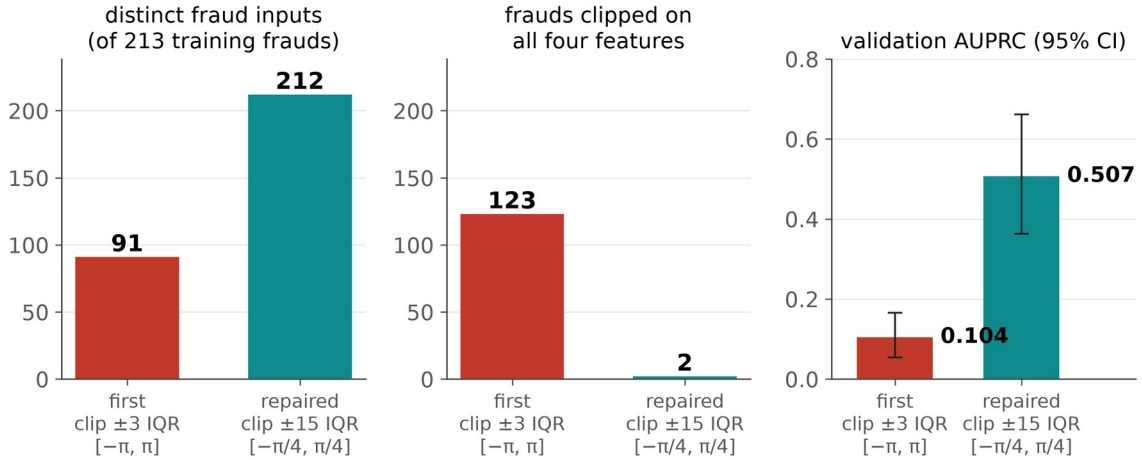
transactions per batch to break even (≈ 80 categorical)

The 66·S probe includes the 2 calibration circuits always-combined already pays, so the break-even is conservative and depends on the gate-free share.

Source: ablation_probes.csv, ablation_probe_cost.csv

B4 · Baselines and the encoding repair





Source: phase2_classical_metrics.csv, p3ar_validation_comparison.csv, p3ar_encoding_diagnostics.csv

B5 · Statistical details

Comparator (5C test)	Material: R_thr − comp.	Any disagreement: R_thr − comp.	Material, validation → test
Gradient boosting	−0.0005 [−0.027, +0.016]	+0.154 [+0.028, +0.391]	0.107 → 0.021
Tree, depth 3	−0.006 [−0.026, +0.005]	+0.133 [+0.003, +0.387]	0.175 → 0.026
GBT benefit model	−0.006 [−0.029, +0.011]	+0.069 [+0.007, +0.173]	0.037 → 0.026
Logistic regression	−0.074 [−0.213, +0.011]	+0.046 [−0.023, +0.152]	0.065 → 0.094
R_obs (rule + readout type)	−0.001 [−0.007, +0.002]	+0.130 [+0.001, +0.384]	—
R_thr (deployed rule)	—	—	**0.020 → 0.020**

5B categorical test: R_cat 0.064 vs R_thr 0.088 (paired 0.024 [0.011, 0.037]); learned comparators 0.118–0.439.

Intervals: two-stage cluster bootstrap over only 4 held-out structures (likely too narrow; always-A3's is degenerate).

Source: final_paired_vs_rthr_5c_test.csv, final_paired_vs_rthr_5b_test.csv, p7_policies_5c_val.csv, final_policies_5c_test.csv

B6 · References

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Source: deliverables/paper/references.bib, deliverables/paper/REFERENCES_VERIFICATION.md